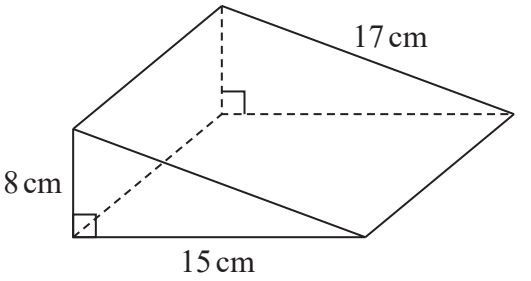
25 The diagram shows a triangular prism.



The total surface area of the prism is 600 cm^2

Work out the volume of the prism.

Area of one triangular end = ½ × 15 × 8 = 60 cm^2

1 mark
The right angle gives perpendicular base and height.

Let the prism length be L cm.

2 × 60 + 8L + 15L + 17L = 600

1 mark
Two triangular ends and three rectangular side faces.

120 + 40L = 600

40L = 480, so L = 12 cm

1 mark
Subtract the two ends, then divide by the triangle perimeter.

Volume = cross-sectional area × length

= 60 × 12 = 720 cm^3

2 marks total
Use volume, not the surface-area formula, for the final step.

720 cm^3

(Total for Question 25 is 5 marks)

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